Project Report

on

Multi-Agent Manufacturing System

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

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Project Number: AAI-32

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Academic Year: 2025-26

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Problem Statement & Objectives

1. Problem Statement

The process of supplier sourcing in manufacturing is often manual, time-consuming, and involves navigating through vast, unorganized datasets. There is a need for an automated system that can efficiently extract relevant supplier information and transform raw manufacturing data into structured, actionable reports.

2. Project Objectives

- To develop an **Agentic AI-based application** that automates supplier sourcing and report generation.
- To implement a **multi-agent collaborative framework** where specialized agents handle distinct tasks like research and writing.
- To utilize **vector databases and semantic search** for fast and accurate retrieval of manufacturing data.
- To provide a user-friendly interface for stakeholders to query sourcing information and receive formatted outputs.

3. Scope of the Project

The project encompasses the design and development of a **Multi-Agent Manufacturing System**. The scope includes:

- Building a modular architecture using **Crew AI** and **LangChain**.
- Integrating **ChromaDB** for vector storage of supplier datasets.
- Developing specialized **Researcher** and **Writer** agents.
- Creating a **Streamlit-based GUI** for user interaction.

Proposed Solution

1. Key Features

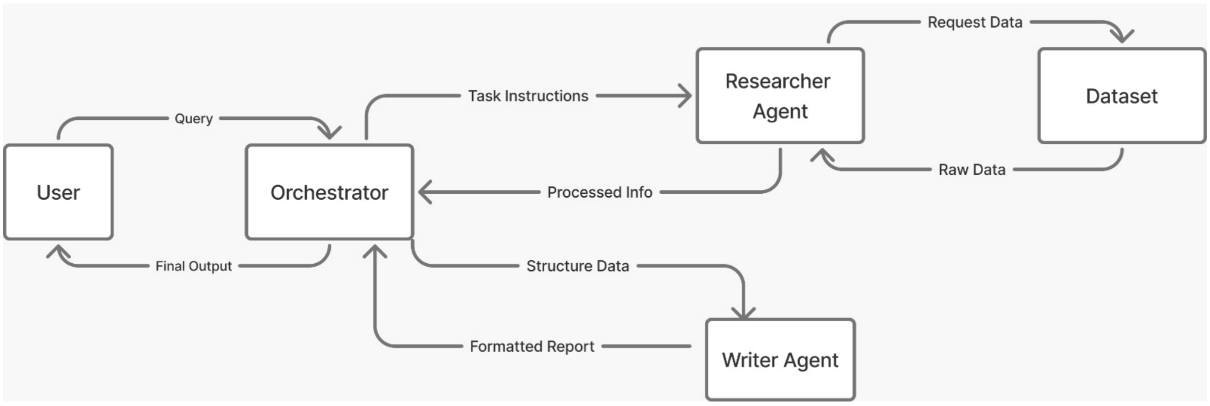
- **Intelligent Orchestration:** Managed workflow and communication between agents.
- **Automated Sourcing:** AI-driven extraction of supplier data from datasets.

- **Structured Reporting:** Automatic formatting of raw data into high-quality, structured reports.
- **Semantic Retrieval:** Context-aware data searching via embeddings.

2. Overall Architecture / Workflow

The system follows an **Orchestrator-Agent pattern**:

1. **User Input:** The user provides a supplier sourcing query through the Streamlit UI.
2. **Research Phase:** The **Orchestrator** assigns the task to the **Researcher Agent**. This agent queries **ChromaDB** using **LangChain** embeddings to extract relevant data.
3. **Writing Phase:** Extracted data is passed back through the Orchestrator to the **Writer Agent**.
4. **Final Output:** The Writer Agent uses the **Gemini LLM** to format a structured report, which is then delivered to the user.



3. Tools & Technologies Used

Component	**Technology**
Agent Framework	Crew AI

Component	Technology
LLM Inference	Google Gemini
Embeddings & Pipeline	LangChain
Vector Database	ChromaDB
User Interface	Streamlit

Results & Output

1. Screenshots / Outputs

The system features a web-based **"Manufacturing Supplier Finder"** interface. Users can enter service requirements into a text input field and click **"Generate Report"** to trigger the multi-agent workflow.

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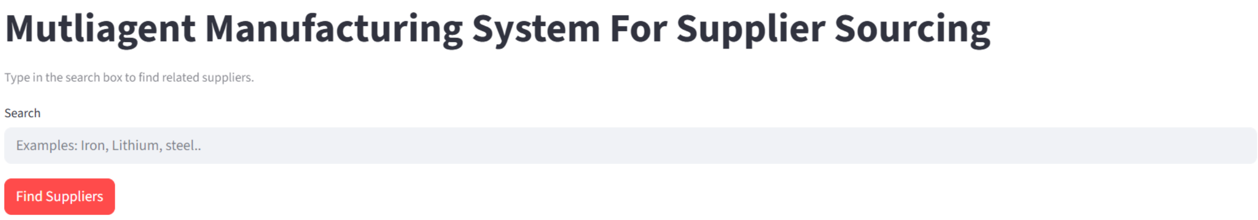
Mutliagent Manufacturing System For Supplier Sourcing

Type in the search box to find related suppliers.

Search

Examples: Iron, Lithium, steel..

Find Suppliers

Home Page


Mutliagent Manufacturing System For Supplier Sourcing

Type in the search box to find related suppliers.

Search

Metals suppliers

Find Suppliers

Found 10 supplier matches from ChromaDB.

Supplier_Name	Material_Type	Material_Name	Location	Lead_Time_Days	Cost_per_Unit_USD	Reliability_Score	Contact_Email	Similarity
Apex Nickel Supplies 27	Metal	Nickel	Delhi	16	309.12	4.88	contact27@supplier.com	0.615
Apex Iron Supplies 194	Metal	Iron	Delhi	27	481.07	4.06	contact194@supplier.com	0.613
Apex Steel Supplies 163	Metal	Steel	Hyderabad	17	384.18	3.71	contact163@supplier.com	0.611
National Copper Supplies 32	Metal	Copper	Mumbai	21	106.24	3.02	contact32@supplier.com	0.603
Advanced Nickel Supplies 18	Metal	Nickel	Delhi	28	410.5	4.46	contact18@supplier.com	0.602
Apex Nickel Supplies 138	Metal	Nickel	Hyderabad	17	204.45	3.18	contact138@supplier.com	0.598

Supplier_Name	Material_Type	Material_Name	Location	Lead_Time_Days	Cost_per_Unit_USD	Reliability_Score	Contact_Email	Similarity
Apex Nickel Supplies 27	Metal	Nickel	Delhi	16	309.12	4.88	contact27@supplier.com	0.615
Apex Iron Supplies 194	Metal	Iron	Delhi	27	481.07	4.06	contact194@supplier.com	0.613
Apex Steel Supplies 163	Metal	Steel	Hyderabad	17	384.18	3.71	contact163@supplier.com	0.611
National Copper Supplies 32	Metal	Copper	Mumbai	21	106.24	3.02	contact32@supplier.com	0.603
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Apex Nickel Supplies 138	Metal	Nickel	Hyderabad	17	204.45	3.18	contact138@supplier.com	0.598
Advanced Nickel Supplies 112	Metal	Nickel	Hyderabad	4	146.76	3.55	contact112@supplier.com	0.596
Advanced Cobalt Supplies 159	Metal	Cobalt	Hyderabad	25	389.51	4.19	contact159@supplier.com	0.595
Advanced Copper Supplies 10	Metal	Copper	Hyderabad	5	413.53	3.3	contact10@supplier.com	0.595
Advanced Copper Supplies 174	Metal	Copper	Ahmedabad	10	238.22	4.17	contact174@supplier.com	0.590

Generated Output for Query

2. Reports / Dashboards / Models

- **Data Model:** Operates on a structured JSON-based retrieval model containing the query, supplier data list, processed info, and the final report string.

  The system operates on a retrieval-based structured data model:

  JSON

  {

  "query": "string",

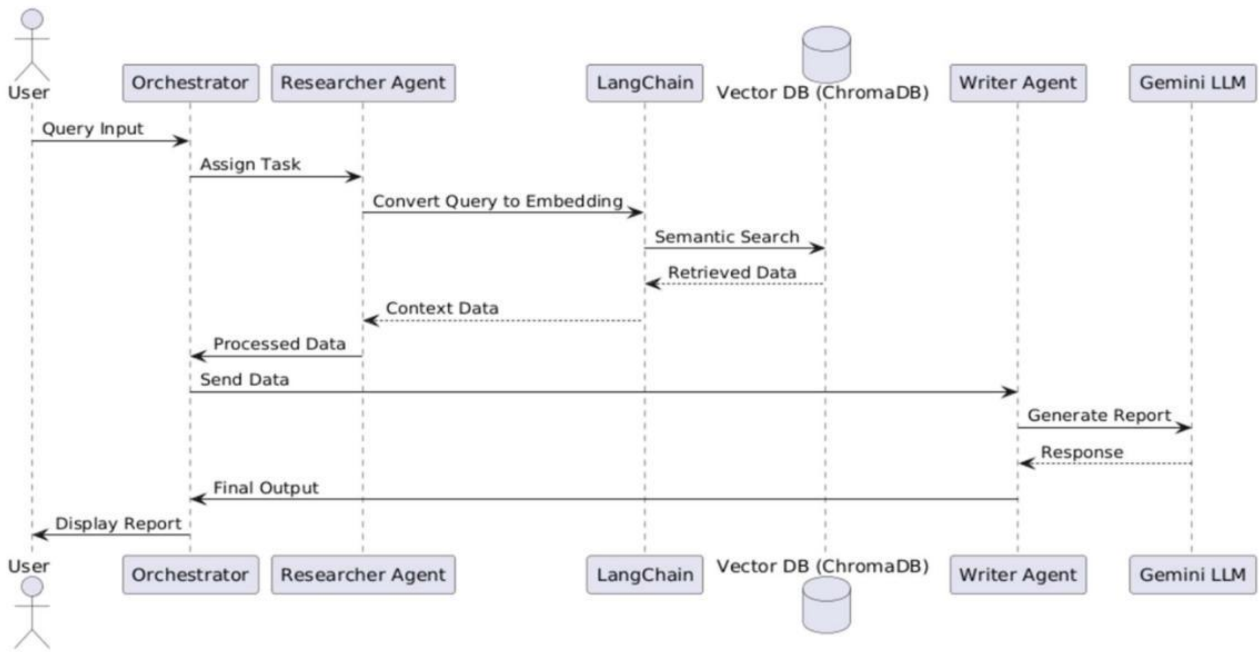
  "supplier_data": ["list"],

  "processed_info": "string",

  "final_report": "string"

  }

- **Workflow State:** Managed by the Orchestrator to ensure seamless hand-offs and consistent state across agents.



3. Key Outcomes

- Successful automation of the end-to-end sourcing process.
- Improved accuracy through **agent specialization** and separation of concerns.

- Efficient processing of manufacturing datasets using semantic search and retrieval.

Conclusion

This project successfully demonstrates the power of **Agentic AI** in the manufacturing domain by creating a **Multi-Agent Manufacturing System**. By orchestrating specialized agents and leveraging vector databases, the system transitions from manual data searching to automated, structured report generation. Key learnings include the effectiveness of **modular agent design** for scalability , the efficiency of **semantic retrieval** for domain-specific data , and the critical role of an **orchestrator** in maintaining system state during complex workflows.

Future Scope & Enhancements

- **Scaling Capabilities:** Adding new specialized agents to handle additional manufacturing capabilities.
- **Permanent Persistence:** Moving beyond volatile session state (which currently clears on refresh) to a persistent database for long-term data retention.
- **Advanced Caching:** Implementing @st.cache_data to optimize performance, as the current version relies on manual session state caching.
- **Extended API Integration:** Further leveraging external APIs for broader reasoning and real-time data processing.